AI Deep Research Module Architecture

This document describes the architecture of the Deep Research Module, a component separate from the document processing pipeline. The Deep Research Module enables general knowledge exploration, advanced prompt-based reasoning, and web-assisted research independent of uploaded documents.

1. Purpose of the Deep Research Module

- Enable general AI-powered research beyond uploaded documents.
- Provide detailed analytical answers to complex prompts.
- Support reasoning, comparison, and knowledge synthesis.
- Optional integration with web search for up-to-date information.

2. Architecture Overview

- User selects Research Mode.
- System bypasses Vector Database (no document retrieval).
- Prompt is sent directly to advanced LLM.
- Optional web search augmentation layer.
- LLM generates structured and analytical output.

3. Core Technologies

- Primary Model: GPT-4o or GPT-4.1 (advanced reasoning).
- Optional Web Browsing Integration for real-time data.
- Prompt Engineering Layer for structured research outputs.

4. Prompt Engineering Strategy

- Use role-based system prompts (e.g., 'You are an expert analyst').
- Encourage step-by-step reasoning.
- Request structured outputs (tables, bullet points, comparisons).
- Define expected depth level (summary, detailed, expert-level).

5. Research Capabilities

- Market analysis.
- Technology comparisons.
- Business strategy evaluation.
- Technical concept explanation.
- Trend analysis and forecasting.

6. Optional Web-Augmented Research Flow

- User submits research query.
- System performs web search (if enabled).
- Relevant sources are retrieved.
- LLM synthesizes information from multiple sources.
- Final structured response is generated.

7. Separation from Document RAG System

- Document Mode: Uses embeddings + vector database + retrieval.
- Research Mode: Direct LLM reasoning without document retrieval.
- Ensures modular and scalable system architecture.

8. Backend Infrastructure

- FastAPI endpoint: /research
- LLM API integration.
- Optional web search API integration.
- Response formatting module.

9. Final Research Flow

- User Query → Mode Detection → (Optional Web Search) → LLM Processing → Structured Research Output.